

의공학개론 (공) (MECH205-01)

학기	2026년 2학기
담당교수	송석호
이수구분	교선평수
학점	3.0
강의시간	화13:00-14:30(E7 - L22), 목13:00-14:30(E7 - L22)
강의실	E5 508 (교수실)
소속	기초학부

교과개요

This course provides an introduction to biomedical engineering for undergraduate students through team teaching by four instructors. Topics span wearable and soft robotics for human augmentation and rehabilitation, soft robotic implants and soft bioelectronics for minimally invasive neural interfaces, biomedical microrobots and robotic catheters, and brain-machine interfaces. Throughout the semester, the fundamental engineering principles underlying these biomedical applications will be covered.

교과목표

Understanding the basic concepts of biomedical engineering in a variety of applications

주차별 강의계획

- 1주차: Biomechanics of Human Motion for Wearable Robotics (E. Rho)
- 2주차: Soft Robotic Gloves & Rehabilitation / Vision-based Control (E. Rho)
- 3주차: Lower-limb Wearable Robots / Full-body Exoskeletons (E. Rho)
- 4주차: ML-based Biosignal Processing / Embodied AI & Future Wearables (E. Rho)
- 5주차: The Soft Robotic Implant Paradigm: Concept, Scope, and Roadmap (S. Song)
- 6주차: Human Anatomy and Immune Response (S. Song)
- 7주차: Conventional Bioelectronic Implants: Target Diseases and Invasive Delivery (S. Song)
- 8주차: Mid-Term Period (No Exam)
- 9주차: Soft Robotic Technologies for Minimally Invasive Implants (S. Song)
- 10주차: Scaling and the Physics of the Microscale; Bioinspired Microrobot (M. Kim)
- 11주차: Microfabrication, Smart Materials, Sensors, and Actuators; Biomedical Functions (M. Kim)
- 12주차: Robotic Catheter and Biomedical Microrobots (M. Kim)
- 13주차: Introduction to Brain-Machine Interface / Neural Signals and Nervous System (A.-H. Lee)
- 14주차: How BMI Hardware Works / Circuits, Electrodes, and Signal Processing (A.-H. Lee)
- 15주차: Advanced BMI Systems (A.-H. Lee)
- 16주차: Final Period (No Exam)

수업방법

Lecture

학습윤리

No distribution of course materials without consent of the instructor. No cheating for all the course activities such as attendance, assignments, and exams.

교과정책(방침)

Attendance is highly recommended. Active participation in the lecture is encouraged.

준비물 및 기타

Lecture materials to be distributed